

Adaptive Experimentation in Education

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Goals

Increase rigorous experimentation by allowing rapid creation and deployment of experiments in educational technologies.

Enable research into adaptive experimentation by providing a testbed for algorithms that dynamically analyze data from experiments, to accelerate improvement with fewer participants

UpGrade

UpGrade (www.upgradeplatform.org) is open source software created by Carnegie Learning to allow the easy creation of experiments by education researchers.

Implemented Systems

EASI is designed as an open API. We have piloted with two EdTech systems, one in Higher Education and one in K-12 Education.

Open Learning Initiative (Higher Ed)

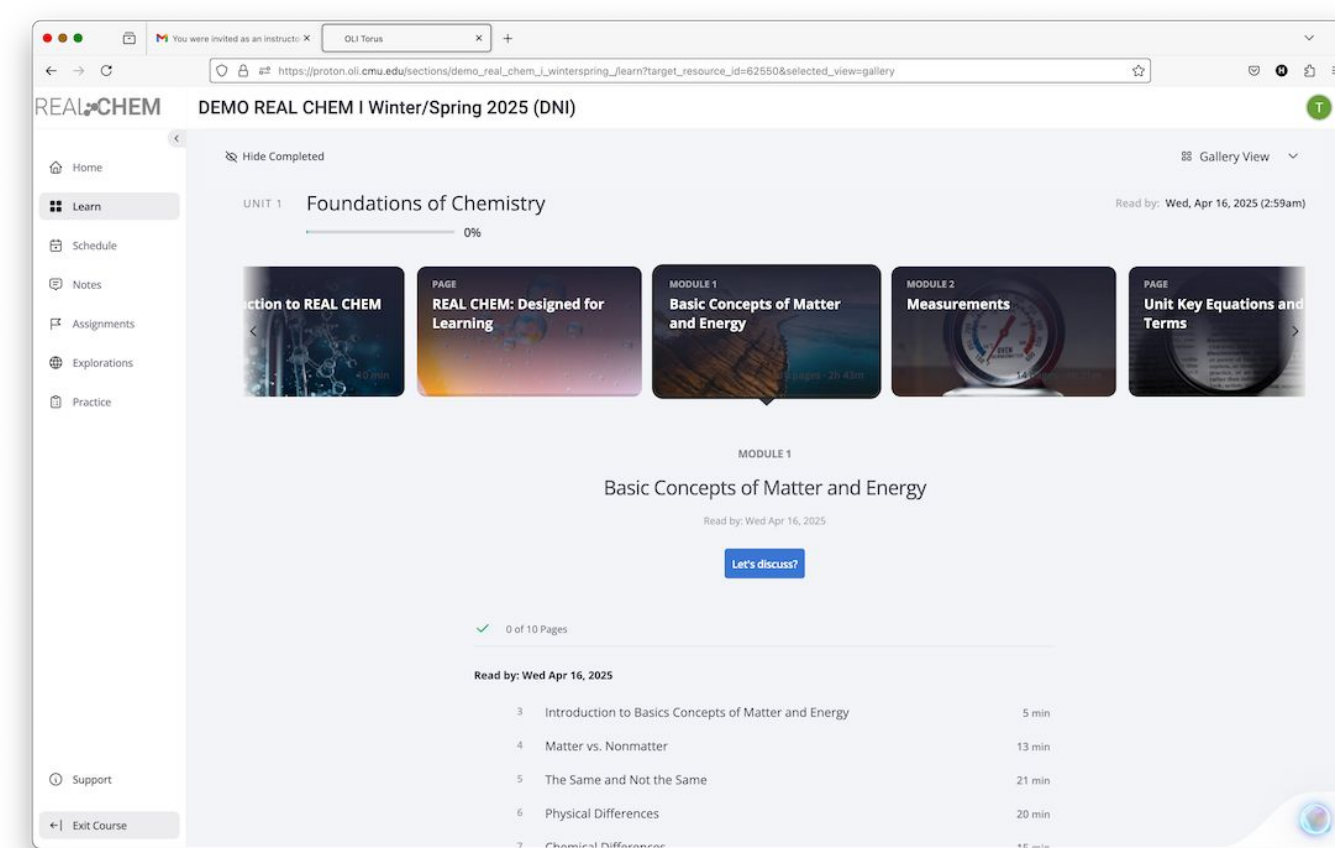
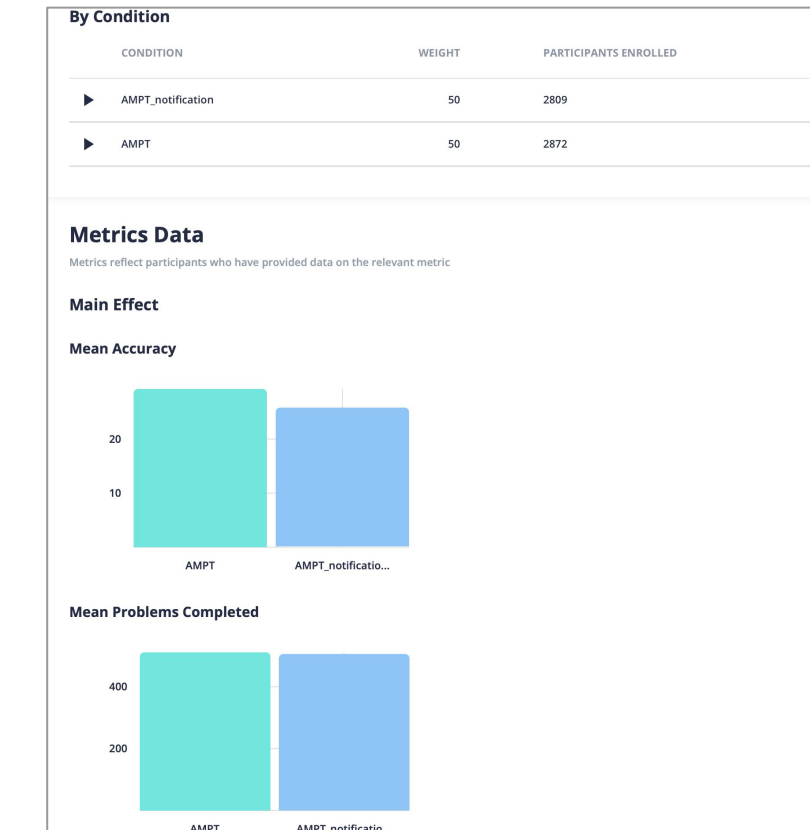
OLI Torus is the newest iteration of a decades-old system for deploying courseware by our partners at Carnegie Mellon. OLI has rich data collection and robust learning activities. OLI is used by hundreds of universities and millions of students.

Adaptive Experimentation

Uses statistical and machine learning techniques to adjust condition probabilities and dynamically deploy results to support rapid learning improvements. The core system is based on the MOOClet Framework developed at the University of Toronto.

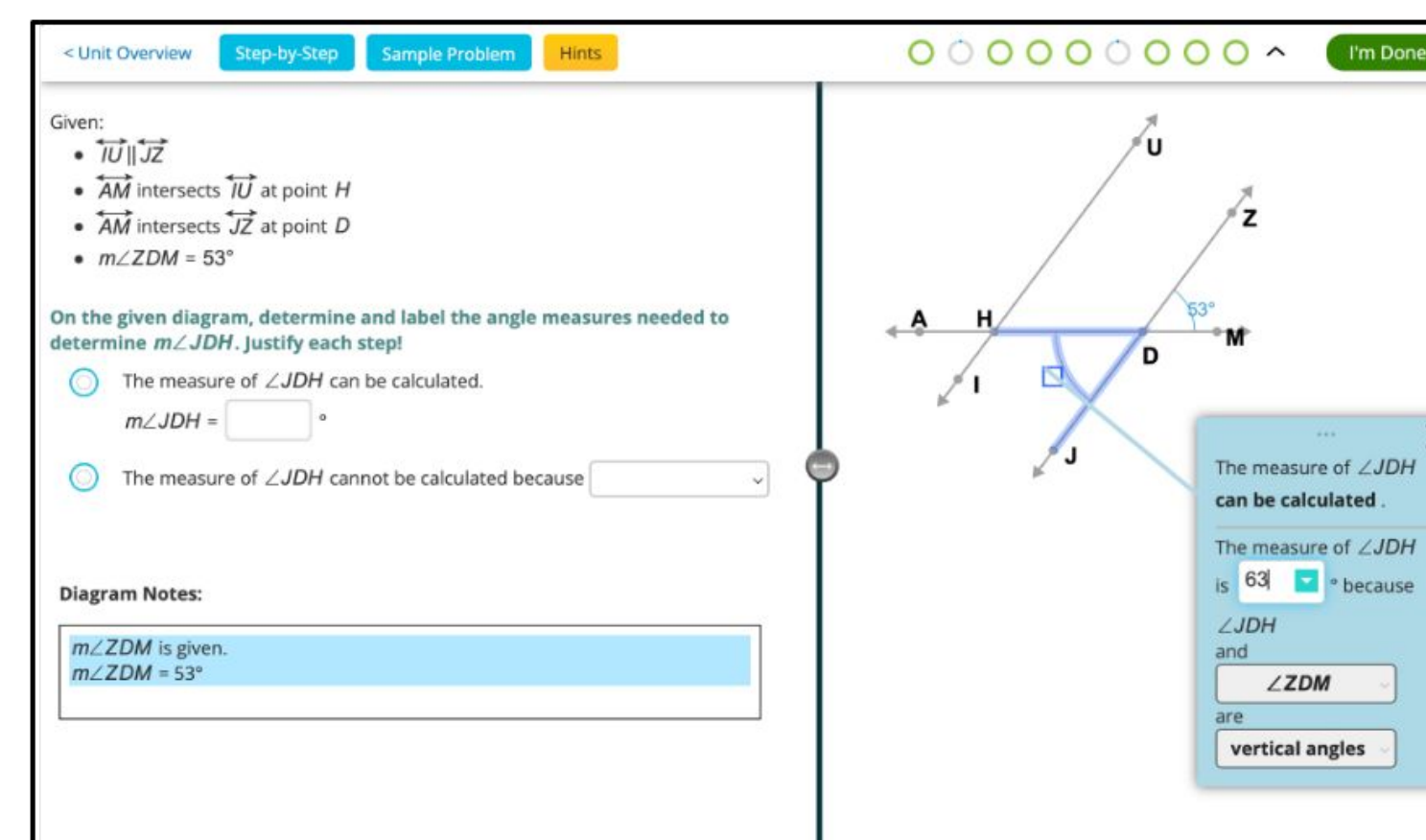


Adaptive Experiments in UpGrade



MATHia (K-12)

MATHia is Carnegie Learning's flagship application. Used by more than 700,000 middle and high school students, the software is based on cognitive tutor technologies originally developed at Carnegie Mellon.



Next Steps

Adaptive experimentation is particularly useful when there is a large number of possible conditions and where outcome measures can be immediately collected.

AI-based chatbots, where students rate their interactions seems like a good application. We will be piloting EASI in this context with students in the Fall.



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